

# Joshua Elieson

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## Skills

**Languages:** JavaScript, TypeScript, Java, Python, C/C++, C#, SQL, MATLAB

**Technologies & Tools:** AWS, GCP, Docker, Kubernetes, Terraform, Linux, Bash, Node.js, React, RESTful APIs, PostgreSQL, Git, GitHub, Jenkins, Jira, Agile, VSCode, Visual Studio

**Clearance:** U.S. Citizen with active U.S. Government Security Clearance (Secret)

## Education

**Columbia University**

*M.S. in Computer Science*

Aug 2026 – Dec 2027

*Incoming Fall 2026*

**University of Utah**

*B.S. in Computer Engineering*

Aug 2022 – May 2026

*GPA: 3.84/4.0*

## Work Experience

**Northrop Grumman**

Aug 2024 – Aug 2025

**Software Engineering Co-op (Cloud Services and Integration)**

*Roy, UT*

- Developed 10+ automated validation scripts in Jython and Groovy to scan thousands of UML system models for security and deployment errors. Reduced manual review time by 60% and prevented critical integration failures before release.
- Refactored and optimized a complex CI/CD pipeline, cutting deployment time from 8+ hours to under 10 minutes. Supported faster delivery cycles for 200+ internal projects.
- Built and maintained microservice components using enterprise patterns common in AWS and Java-based environments, improving system reliability and supporting seamless deployment across distributed teams.
- *AWS, Java, Jython, Groovy, C++, Git, CI/CD, Agile*

**Northrop Grumman**

May 2024 – Aug 2024

**Software Engineering Intern (Physical Security)**

*Roy, UT*

- Implemented back-end linear algebra algorithms in C to convert live incoming raw LIDAR data packets into structured point cloud (PCD) files, improving processing speed by 3× and delivering a smoother visualization for security and other developers.
- Developed front-end features in React to render interactive 3D point clouds, enabling non-technical users from 5+ internal teams to quickly analyze sensor data directly in a web application.
- *TypeScript, React, JavaScript, C, Node.js, REST APIs, Git, Linux, Jira, Agile*

**University of Utah**

Aug 2023 – May 2024

**Teaching Assistant (Physics for Scientists and Engineers)**

*Salt Lake City, UT*

- Led bi-weekly lab sessions and office hours for over 100 students, providing individualized support and fostering deeper engagement with core physics concepts throughout the semester.
- Designed and delivered over 20 interactive workshops and demonstrations that simplified complex topics like mechanics and electromagnetism, resulting in a 10% increase in final course grades among students.

## Relevant Projects

- **Mechanical Detection of Magnetic Resonance — Undergraduate Research (2023 – 2025):** Designed and implemented software in Python and MATLAB to process, log, and visualize voltage signals from a custom magnetic-sensing device, achieving a 10× improvement in detection sensitivity from the initial design and demonstrating potential for next-generation magnetic sensors and spin-detection technologies.  
*Python, MATLAB, Signal Processing*
- **Spreadsheet Application — Spring 2025:** Built a C#-based spreadsheet application with dynamic cell dependencies, a custom formula parser and evaluator supporting arithmetic expressions and cell references, and stack-based undo/redox functionality. Designed a responsive interface for real-time updates and intuitive data manipulation.  
*C#, .NET, WinForms, GitHub*
- **Multiplayer Agar.io Clone — Fall 2024:** Built a real-time multiplayer game with a client-server architecture using TCP sockets and HTTP endpoints to manage bidirectional communication. Implemented asynchronous threading and database integration to support concurrent players, handle collisions, and broadcast updates with minimal latency.  
*C#, .NET, Sockets, Networking, HTTP, SQL*